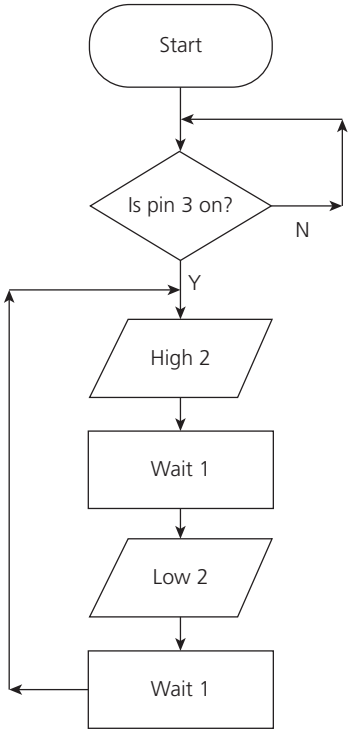


Figure 1.23 A program that uses time delays and loops to create a continuous pulsing sequence

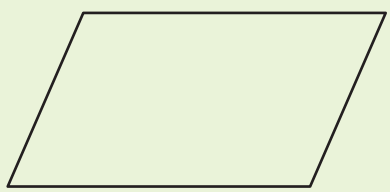
Exam tip

Make sure you can create flowchart programs that can read inputs, switch outputs on/off and produce more complex process functions, such as time delays and counting.

Now test yourself

TESTED

1 What does the following flowchart symbol represent? (1)



2 Draw the flowchart symbol for a process command. (1)

3 Explain how a microcontroller could be programmed to read an analogue value from a sensor. (3)

4 Write a microcontroller flowchart program that would turn an output connected to pin 4 'on' for two seconds when a switch connected to pin 3 is pressed, and then 'off' again. (6)

1.8 Ferrous and non-ferrous metals

1.8.1 Ferrous metals

REVISED

- **Ferrous metals** contain iron so they corrode quickly.
- They are magnetic and corrode quickly if not treated with a suitable surface finish.

Ferrous metals: metals that contain iron.

Mild steel

- Mild steel is also known as low-carbon steel because it has the lowest carbon content of the different steel types.
- Mild steel contains around 99 per cent iron and a maximum of 0.25 per cent carbon.

- Other elements such as manganese, silicon and sulphur make up the remaining fraction.
- Mild steel is relatively affordable compared to other types of steel.
- It is easy to machine and weld.
- Common uses are structural steel in buildings, signs, car and vehicle bodies and panels, and wire.

Exam tip

Remember that all steels are actually alloys as they are a mixture of iron, carbon and other elements.

Stainless steel

- Stainless steel contains iron and between 15 and 30 per cent of other metals (chromium and nickel).
- It has a high resistance to corrosion compared to other types of steel.
- It is tough and ductile with high strength and temperature resistance.
- Common uses are medical instruments, kitchen utensils and appliances, vehicle exhaust systems, hardware and structural/architectural items.

Cast iron

- Cast iron is a mixture of iron and two per cent or more of carbon.
- It is heavy and has good compressive strength but is quite **brittle** and can crack or shatter under impact.
- It has a low melting point and is more resistant to wear and corrosion than mild steel.
- Common uses are pipework, engines, cooking utensils, and garden ornaments and furniture.

Brittleness: hard but easily broken or shattered.

1.8.2 Non-ferrous metals

REVISED

- **Non-ferrous metals** do not contain iron.
- They are much more corrosion resistant than ferrous metals.
- They are generally more expensive than ferrous metals.

Non-ferrous metals: metals that do not contain iron.

Aluminium

- Aluminium has a high strength-to-weight ratio as it is lightweight.
- It has a shiny, attractive appearance that requires no additional finish.
- It is easy to cut, shape and weld.
- Common uses are drinks cans, foils, window frames and aeroplane parts.

Copper

- Copper is an excellent electrical and thermal conductor that does not tarnish or corrode easily.
- It is tough but ductile, so it can be easily shaped or drawn into a wire.
- Copper can be joined easily by soldering or brazing.
- It can be polished to an attractive appearance and golden colour.
- Common uses are water pipes, electrical cables, jewellery, statues and outdoor decorative features.

Brass

- Brass is an alloy of copper and zinc.
- Alloys are metals made by combining two or more metals and other elements together to improve their properties.
- Brass has good resistance to wear and corrosion, and has an attractive appearance.
- Common uses for brass are nuts, bolts and screws, door furniture and bathroom fittings.



Figure 1.24 Brass is an alloy made from copper and zinc

Typical mistake

Unlike gold and silver which are non-ferrous metals, brass is an alloy and not a 'pure' metal.

1.8.3 Properties

REVISED

The properties of different metals are the main factors that will influence their selection and use. The general properties of metals are:

- **high melting point**: resistant to very high temperatures
- **finish**: can be polished to an attractive shine
- **conductive**: allows electricity and heat to pass through easily
- **malleable**: can be hammered and bent into shape without breaking
- **ductile**: can be stretched into a thin wire without becoming weaker or brittle in the process
- **hardness**: resistant to impact and abrasion.

Conductive: allows electricity and heat to pass through easily.

Malleable: can be hammered and bent into shape without breaking.

Ductile: can be stretched into a thin wire without becoming weaker or brittle in the process.

Now test yourself

TESTED

1 Name the **two** differences between ferrous and non-ferrous metals. (2)
2 What is an alloy? (2)
3 All metals are good conductors. What does this mean? (2)

1.9 Papers and boards

1.9.1 Paper

REVISED

- Papers and boards come in a wide range of different thicknesses, sizes and types.
- The thickness of paper is known as its 'weight' and is measured in **grams per square metre (gsm)**.

Grams per square metre (gsm): the weight in grams of one square metre of paper.

Copier paper

- Copier paper is used extensively for printing, photocopying and general office purposes.
- It is approximately 80 gsm.
- It has a smooth surface making it ideal for most printers and photocopiers.

Exam tip

Papers heavier than 170 gsm are classified as boards.

Cartridge paper

- Cartridge paper is available in different weights between 80 and 140 gsm.
- It is thicker and more expensive than layout and copier paper.
- It has a slightly textured surface and is slightly creamier in colour.
- Cartridge paper is used by artists for sketching, drawing and painting as it has an ideal surface for pencil, crayons, pastels, watercolour paints, inks and gouache.

Tracing paper

- Tracing paper is used for making copies of drawings and fine details.
- It is thin and very transparent.
- It is also hardwearing and strong despite its lack of thickness.
- Mistakes made in pen can be scratched off using a sharp blade.

1.9.2 Board

REVISED

- Boards are usually classified by thickness as well as by weight.
- The thickness of board is measured in **microns**; a micron is one-thousandth of a millimetre.

Folding boxboard

- Folding boxboard is made up of three layers of card.
- The card layers have different qualities and properties: the two outer layers are high quality with a very smooth surface; the middle layer is lower quality but has high bulk and **stiffness**.
- A chemical coating on the top outer layer makes it ideal for high-quality printing and foiling.
- Folding boxboard is used for health and beauty products, frozen and chilled foods, pharmaceutical products and confectionaries.

Stiffness: resistance to bending and flexing.

Corrugated cardboard

- Corrugated cardboard is a strong but lightweight type of card.
- It is made from two layers of card with a fluted sheet in between.
- The fluted construction makes it stiff and difficult to bend or fold, especially across the flutes.
- Corrugated card can absorb knocks and bumps, and has good heat-insulating properties.
- Common uses are packaging fragile or delicate items, and takeaway foods, for example pizza boxes.

Solid white board

- Solid white board (also known as mounting board) is a rigid type of card with a thickness of around 1.4 mm (1400 microns) and a smooth surface.
- It is available in different colours but white and black are the most commonly available.
- Mounting board is often used for picture framing mounts and architectural modelling.